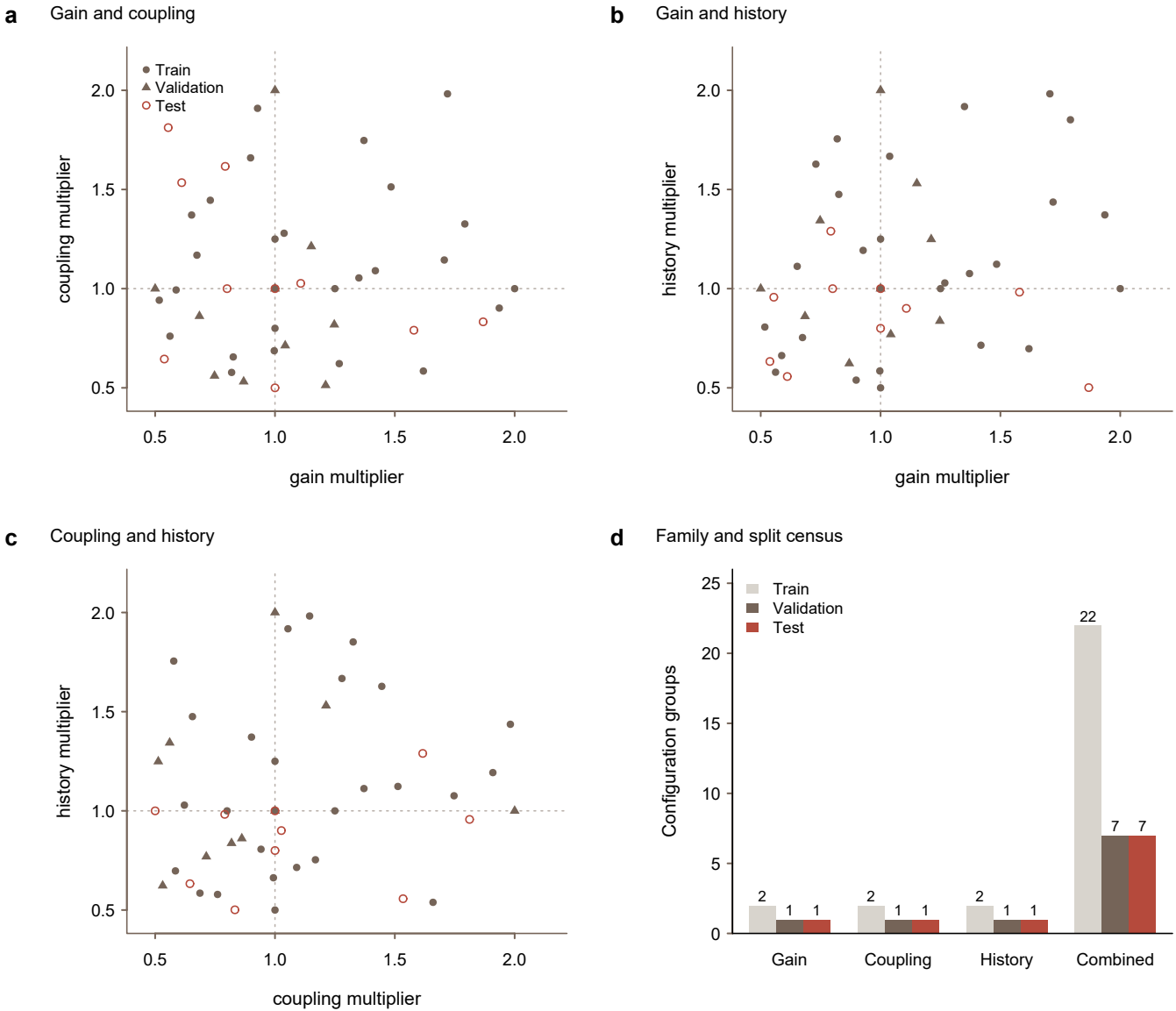
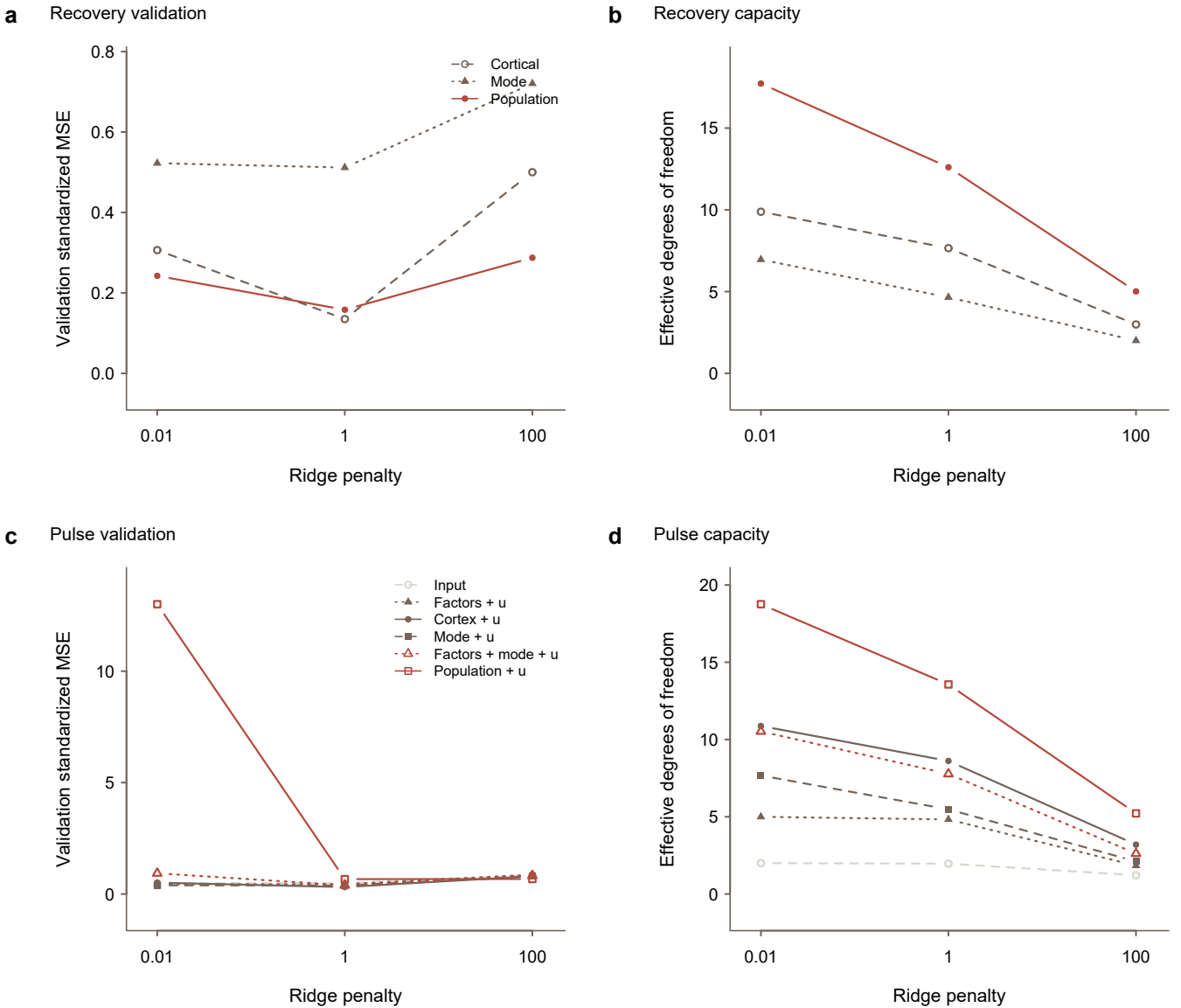
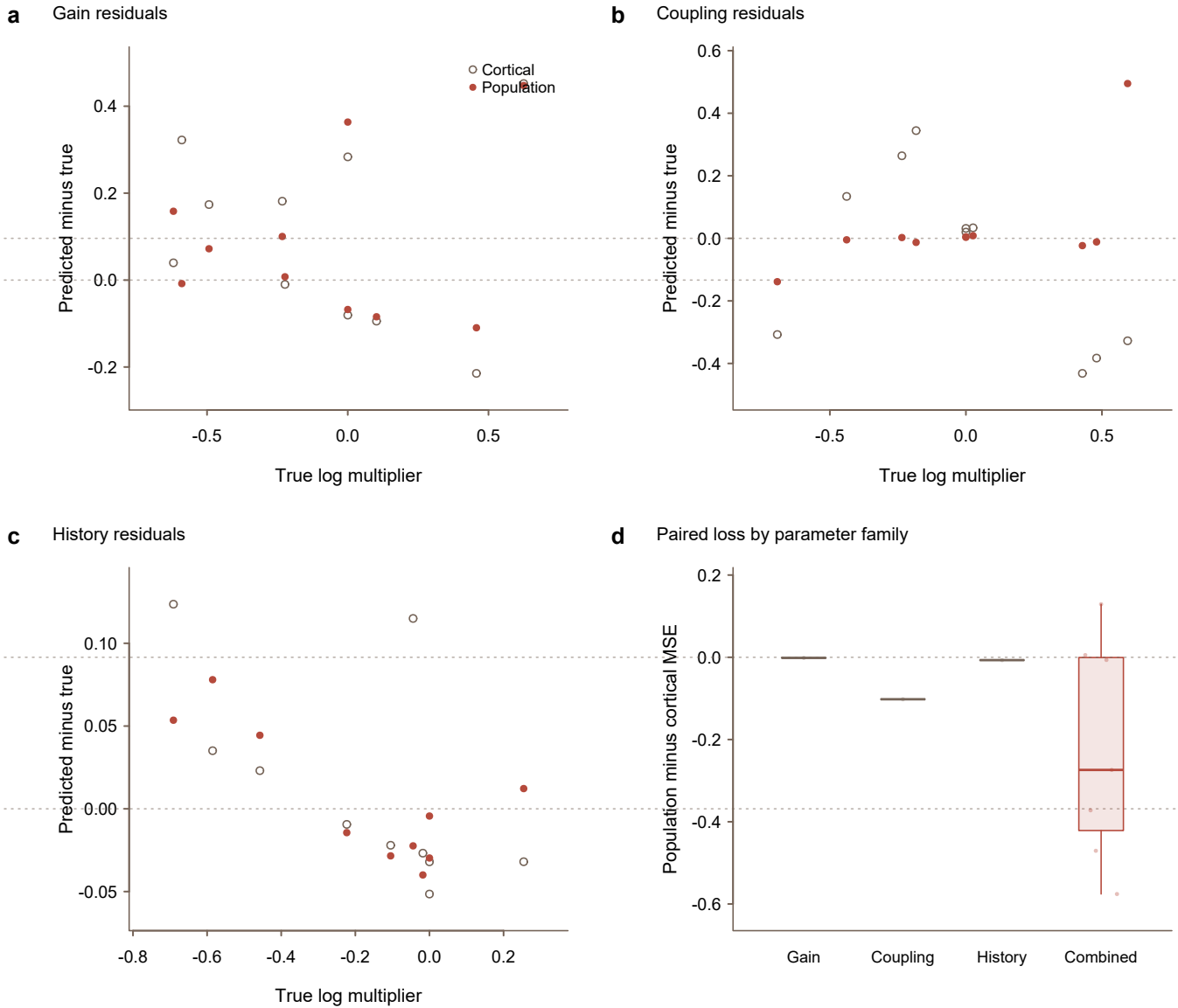
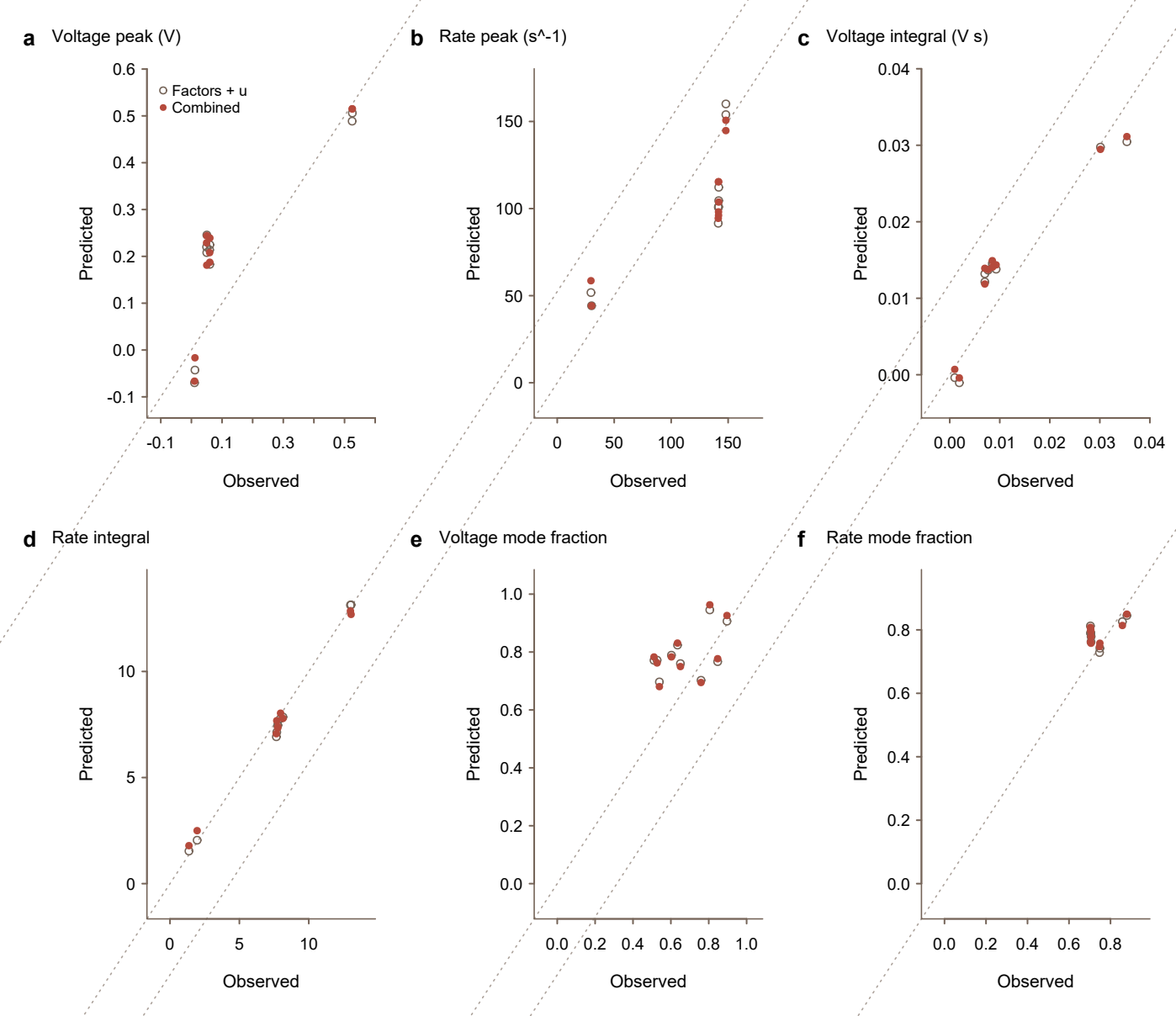
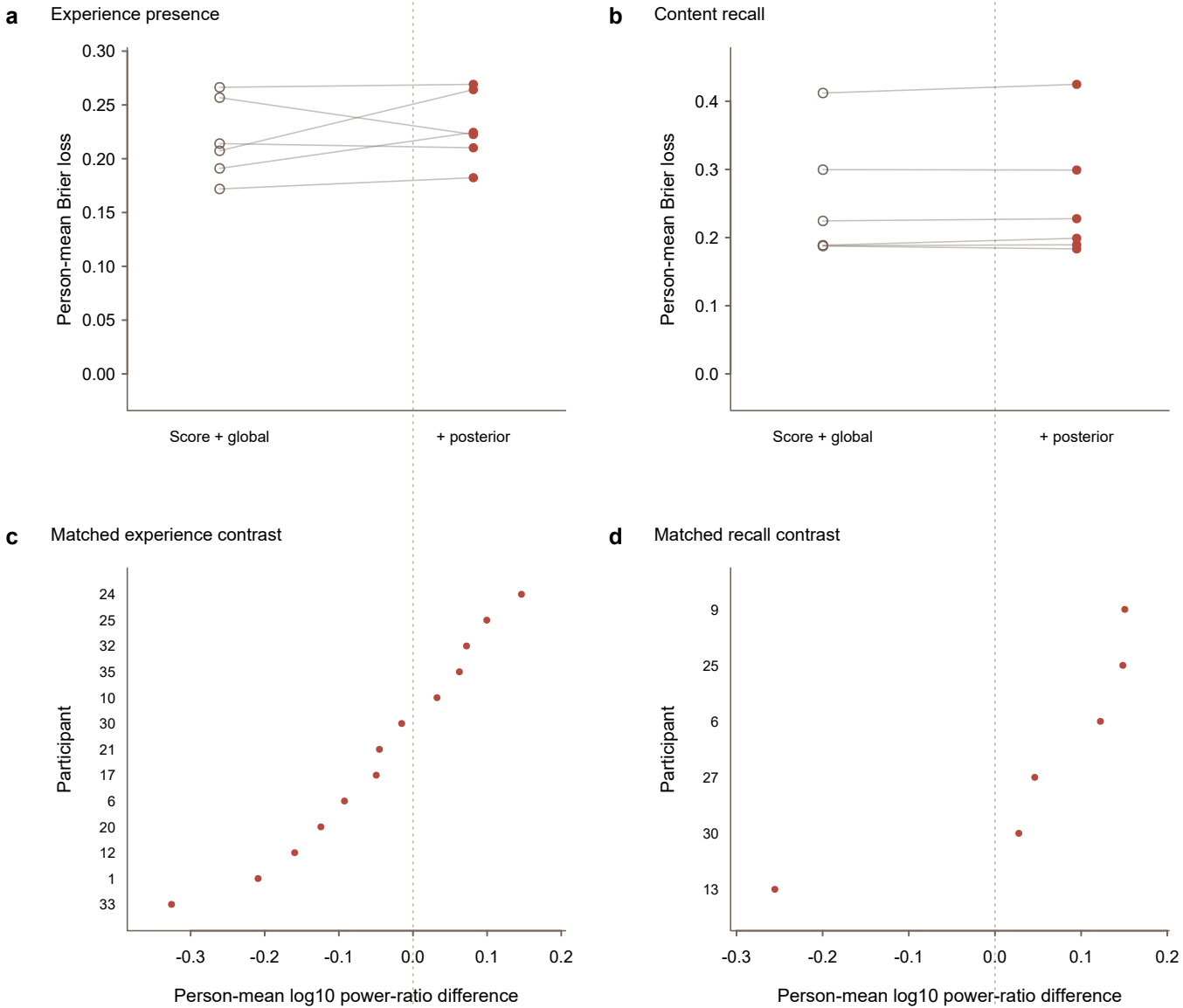
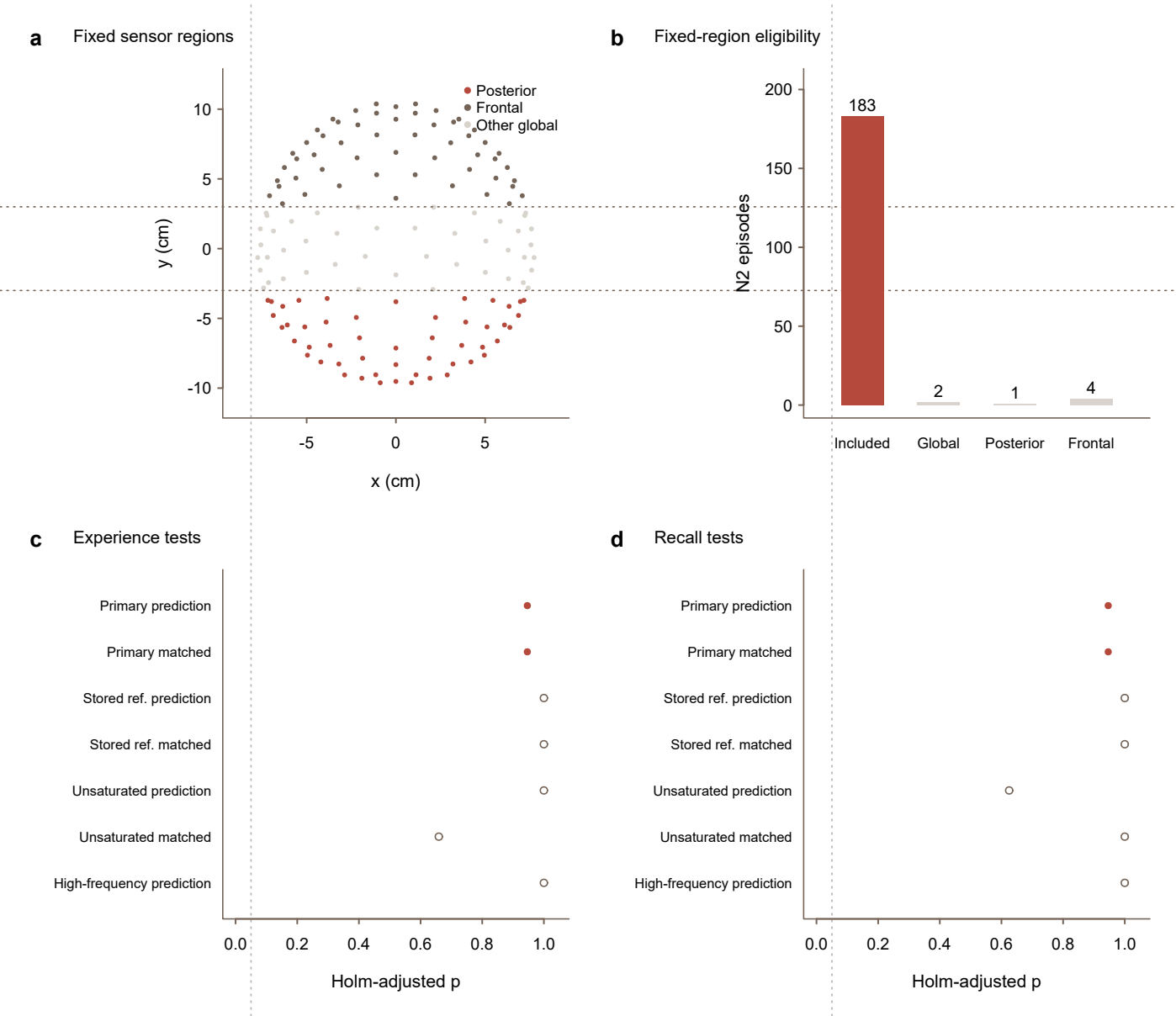

a
Gain and coupling
Train
Validation
Test
coupling multiplier
gain multiplier
b
Gain and history
history multiplier
gain multiplier
c
Coupling and history
history multiplier
coupling multiplier
d
Family and split census
Train
Validation
Test
Configuration groups
Gain
Coupling
History
Combined
2
1
1
2
1
1
2
1
1
22
7
7

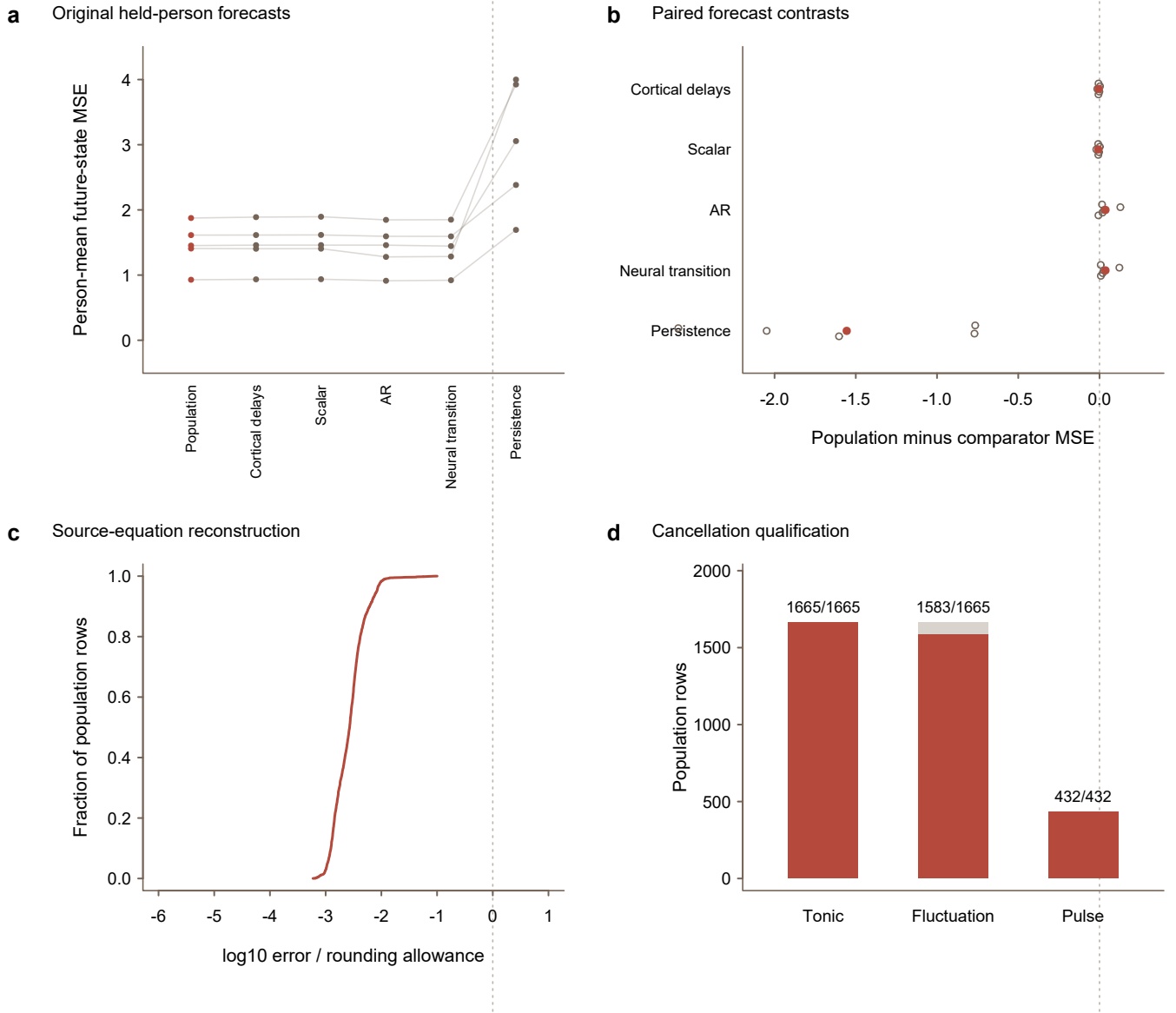

a
Recovery validation
Validation standardized MSE
0.8
0.6
0.4
0.2
0.0
Cortical
Mode
Population
0.01
1
100
Ridge penalty
b
Recovery capacity
Effective degrees of freedom
15
10
5
0
0.01
1
100
Ridge penalty
c
Pulse validation
Validation standardized MSE
10
5
0
Input
Factors + u
Cortex + u
Mode + u
Factors + mode + u
Population + u
0.01
1
100
Ridge penalty
d
Pulse capacity
Effective degrees of freedom
20
15
10
5
0
0.01
1
100
Ridge penalty


a
Gain residuals
Cortical
Population
Predicted minus true
True log multiplier
b
Coupling residuals
Predicted minus true
True log multiplier
c
History residuals
Predicted minus true
True log multiplier
d
Paired loss by parameter family
Population minus cortical MSE
Gain
Coupling
History
Combined


a
Voltage peak (V)
Factors + u
Combined
Predicted
Observed
b
Rate peak (s^-1)
Predicted
Observed
c
Voltage integral (V s)
Predicted
Observed
d
Rate integral
Predicted
Observed
e
Voltage mode fraction
Predicted
Observed
f
Rate mode fraction
Predicted
Observed


a
Experience presence
Person-mean Brier loss
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Score + global
+ posterior
b
Content recall
Person-mean Brier loss
0.4
0.3
0.2
0.1
0.0
Score + global
+ posterior
c
Matched experience contrast
Participant
24
25
32
35
10
30
21
17
6
20
12
1
33
-0.3
-0.2
-0.1
0.0
0.1
0.2
Person-mean log10 power-ratio difference
d
Matched recall contrast
Participant
9
25
6
27
30
13
-0.3
-0.2
-0.1
0.0
0.1
0.2
Person-mean log10 power-ratio difference


a
Fixed sensor regions
Posterior
Frontal
Other global
y (cm)
x (cm)
b
Fixed-region eligibility
N2 episodes
183
2
1
4
Included
Global
Posterior
Frontal
c
Experience tests
Primary prediction
Primary matched
Stored ref. prediction
Stored ref. matched
Unsaturated prediction
Unsaturated matched
High-frequency prediction
Holm-adjusted p
d
Recall tests
Holm-adjusted p


a
Original held-person forecasts
Person-mean future-state MSE
0
1
2
3
4
Population
Cortical delays
Scalar
AR
Neural transition
Persistence
b
Paired forecast contrasts
Cortical delays
Scalar
AR
Neural transition
Persistence
-2.0
-1.5
-1.0
-0.5
0.0
Population minus comparator MSE
c
Source-equation reconstruction
Fraction of population rows
0.0
0.2
0.4
0.6
0.8
1.0
-6
-5
-4
-3
-2
-1
0
1
log10 error / rounding allowance
d
Cancellation qualification
Population rows
0
500
1000
1500
2000
1665/1665
1583/1665
432/432
Tonic
Fluctuation
Pulse